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Privacy-Preserving and Quantum-Resistant Federated Learning Framework for Fog Healthcare Systems

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Abstract

Quantum computing is on the horizon and will destroy existing cryptography standards, putting digital healthcare system security and patient privacy at danger. For very private and secure communication in fog healthcare settings, this article presents a new Quantum-Resistant Federated Deep Learning (QR-FDL) Framework. The QR-FDL architecture incorporates a lattice-based technique (like Kyber) into the model aggregation stage of Federated Learning (FL), which is a kind of Post-Quantum Cryptography (PQC). We provide a solution that uses Differential Privacy (DP) to prevent attacks based on gradients in data reconstruction and guarantees quantum-era security for weights of models with gradients sent between the central server and decentralized fog nodes. The practical use of QR-FDL is shown by its rapid convergence rate and excellent classification accuracy of 91.5% in an empirical assessment including a medical imaging job, such as tumor classification. Critically, we measure the cryptographic overhead and demonstrate that, even if PQC causes a regulated latency rise the total FL round time is still tolerable for real-time fog settings. In order to implement quantum-secure along with privacy-preserving Deep Learning in mission-critical healthcare communications, this study presents the first proven, end-to-end solution. In addition to showing that federated medical facilities can be securely encrypted from end to end, this architecture uses differential privacy during model training to protect patients' personal information. Through the research, we demonstrate its practical implementation in the near future for it is hard to balance the precision of modeled obstructions with cryptographic overhead that still remains acceptable. The results of this research lay the groundwork for medical AI systems of the future that are both secure and respectful of patients' privacy.

Keywords Smart Healthcare · Internet of Things (IoT) · Fog Computing · Federated Deep Learning (FDL) · Quantum Key Distribution (QKD) · Post Quantum Cryptography (PQC) and · Differential Privacy (DP)

1 Introduction

A new and exciting paradigm for distributed machine learning (ML) is federated learning (FL). It shines in fields where patient confidentiality is paramount, including medical [1]. While traditional neural network training methods require consolidation of data at a single training center, FL can enable multiple separate clients to participate



in the training process and collectively form a model. Data localization is a feature that is maintained by each client in FL. Meanwhile, the extent of QC has been increasing at an exponential level [2]. QC does computations using quantum features like superposition and coupling. The new field of quantum ML (QML) was motivated by the idea that quantum technology could surpass classical computing in some ML applications. In particular, the added value of machine learning to quantum computing has garnered the interest of many researchers and universities. This activity has the goal of creating a federated quantum machine learning architecture [3] that will make medical imaging tasks more optimized, safer, and keeps patient's information confidential.

With the tremendous growth of medical information, there is a demand for distributed and comprehensible and secure automated learning systems. To cater these needs, federated learning (FL) has been recently popularized. However, FL models often fail to preserve the privacy of users, to be fair and transparent, and to thwart attacks from adversaries [4]. Protecting sensitive medical information is complex, and poses many challenges for the implementation of a broad application of Deep Learning's potential. Balancing access to medical data for knowledge with privacy is easily achieved. Federated Learning (FL), which allows for distributed model training through collaboration and sharing, emerges as a potential solution. Bypassing data centralization and maintaining data privacy is achieved using this strategy [5]. For example, machine vision and DeepLearning can nowadays identify if a person has skin cancer, through image classification.

Concerns about personal information security have been voiced in relation to centralized learning (CL). This is because there is a limitation on sending large amounts of data to one server [6].

The extremely personal and delicate nature of medical data causes it to be dispersed across many organizations. This is how EHRs are handled solely by hospitals and clinics. Consequently, important data remains isolated within companies, hindering cross-source research that could yield better results [7]. In particular, FL is a relatively new privacy option. It trains local models using data collected locally. In order to enhance generalization characteristics, these local models work together to update a global model. Data corruption and hostile assaults are examples of security vulnerabilities that can compromise computer systems and cause models to produce inaccurate predictions [8].

Recently, FL has gained popularity as a way to train distributed models together without exposing patients' personal health information. But it's still not easy to keep patients' personal information private when they learn and when they share models with a centralized server. A further issue is bias. Another obstacle is the need for fairness in the results of the models. An additional complication is introduced by client heterogeneity. There is also the matter of limited processing to consider [9]. With the growing dependence on AI-driven diagnoses, protecting patient data during medical image categorization has become an urgent issue in the healthcare system. The necessity for safe solutions is underscored by the fact that more than 30% of medical organizations worldwide have suffered a data breach during the past year [10]. These privacy and safety violations continue to occur in autonomous learning environments and are alarming. The need for higher-level security technologies such as quantum-resistant methods to protect sensitive health information is all-important.

Modern healthcare has been utterly transformed by state-of-the-art technology like digital twins (DT), blockchain, and intelligent machines (AI). By developing more accurate diagnostic tools, they have improved the standard of public health and medical care. The enormous volume of biological information still poses significant obstacles, even with recent improvements. Data privacy, scalability, and security are the three main obstacles [11]. Practical situations requiring data sharing and collaboration encounter specific obstacles due to the ever-increasing amount of healthcare and medical data originating from various sources. Security breaches and problems with data fusion are two examples of these difficulties. Weak data exchange and unreliable data storage are two more problems [12].

There is a lot of hope that federated learning can pave the way for massive healthcare studies. It also makes it easier for different centers to work together. Concurrently, it safeguards the confidentiality and integrity of data. Despite the abundance of current research on healthcare that makes use of or suggests FL-based technologies, it is still not apparent which ones could be clinically useful [13]. Privacy concerns and regulations on exchanging

data could limit the medical field's use of this technology. In addition, when a healthcare predictive framework is compromised, it poses a threat to human life. Therefore, strong defences against intrusions are crucial for the healthcare industry [14]. The use of quantum-resistant cryptographic algorithms in the FL mechanism is crucial to ensure that these more sophisticated threats do not compromise the security of the data and that it remains fully intact. However, a benefit is that sensitive clinical information is handled with care and privacy ensured.

The integration of quantum computing with the current power of advanced systems has the potential to revolutionize the medical field. This can increase the data processing speed of data study and facilitate the process of studying data. Its use might also give rise to enhanced methods of development, along with the ability of optimizing the interpretation of complex data from biology [15]. Protecting private patient data across disparate systems, the federated learning (FL) method has recently emerged in fog-based medical facilities. Innovations in quantum mechanics have the potential to simplify the design procedure. The time and money needed to complete the task are reduced as a result [16]. The fundamental idea behind FL is to build trustworthy models by combining information from different datasets in a decentralized way. Its potential for use in healthcare is enormous because of the mountain of data produced by things like imaging services, wearable tech, and medical files. Data gathered from scientific surveys is also useful for making better models [17].

Despite this, shared learning is more likely to be attacked because it is spread out. The federated learning server could be compromised if users contribute fraudulent data. Also, new research has demonstrated that attackers can get data only by looking at parameters. Therefore, present FL paradigms still have a great deal of space for improvement [18]. Assumption number one: updating a model on a server does not compromise user privacy. This is contingent upon the fact that the updates are immune to being “reverse engineered” to deduce details on the confidential training records. Nonetheless, this privacy principle is not valid under many different conditions [19]. The need for FL that protects privacy is highlighted by these issues. Advanced “quantum-resistant” security procedures should also be implemented to fortify systems. This makes data more secure and protects it from powerful inference assaults.

Nowadays, smart applications rely heavily on the IoT. Some examples of this include healthcare, the house, public transit, and classroom instruction. As ML in the IoT continues to grow, fog computing is looking like a great way to enhance cloud computing. To facilitate the storage and processing requirements of these systems, it brings the cloud out to the network's periphery [20]. Fog computing and federated DL are now working together in this combination. The implementation of quantum-resistant techniques further improves it. Their combined strength makes them an ideal framework for creating solutions that are both extremely safe and respectful of users' privacy. Because medical data is both dispersed and delicate, this method handles it right at the outermost layer of the network.

The primary objective of this work is to develop a privacy-preserving and post-quantum-resistant federated learning framework for dementia MRI classification in fog healthcare environments. The proposed system aims to protect sensitive patient information during decentralized model training while maintaining high classification performance and acceptable communication overhead.

2 Related Works

The authors of [21] suggest a two-stage security architecture to improve authentication as well as encryption for fog-IoT devices. The first step combined Elliptic-Curve Diffie-Hellman with the Lorenz chaotic system to ensure that the keys are not predictable in a strong way. The second step used the cryptographically sound CRYSTALS-Kyber method to guard sensitive information. Tests conducted by NIST SP 800 – 22 verified the robust key randomization. The examination of efficiency revealed that, in comparison to previous systems, creating keys, encoding, and decoding were performed more quickly.

Data usable while maintaining privacy, which are given by utility-aware privacy preservation techniques, enhance DP knowledge. By providing quantifiable privacy mechanisms to offer enhanced privacy protection in

location-based services without compromising the prediction accuracy, it is suitable for such intelligent applications as distributed intelligent healthcare applications, which requires patient privacy protection, and secure data sharing environments in fog computing systems [22].

To strengthen the confidentiality and integrity of smart healthcare systems built on the blockchain, a multiple-layered model is proposed in [23]. This system enhances network safety and protects patient confidentiality while allowing for the safe utilization of data related to rare diseases. Federated Learning (FL) and Differential Privacy (DP) are both comprised of it. Optimizing model precision is achieved by the utilization of adaptive noise dispersion. AES and a quantum-resistant mechanism called ML-KEM work together to make sure that transmission is safe.

A grasping mechanism using allies with occlusion was presented for the robotic automation in complex environments. Intelligent learning strategies are used by the system to further enhance the object detection and grasping performance under occluded circumstances and can contribute to securely realize the automation of healthcare service and assistive robotic application system in the unsure medical system with wide-spread network [24].

A multifaceted architecture for safe and extensible health IT operations is presented in the paper [25]. It used Hyperledger Fabric and Holochain DHT to support large amounts of data and update times of less than one second. Signs made with CRYSTALS-Dilithium and Kyber key encapsulation protect quantum computing. To detect anomalies while protecting user privacy, edge-oriented collaborative learning using compact autoencoders is employed. Predicting the need for a colostomy in premature babies with NEC using data from maternal sensors and pictures of the baby's intestines proved the system's viability.

A video question answering (VQA) task was proposed with the assistance of a text-based, spatial & temporal attention network. The contribution has demonstrated the effectiveness of the deep learning architectures for intelligent video analysis and healthcare-based visualization data interpretation tasks by integrating multimodal attention mechanisms, to cope with temporal and spatial relationships conveyed through videos [26].

To improve the safety of Fog-Edge computing, the writers of [27] suggest combining blockchain-based FL and Post-Quantum Privacy. Cryptography built around multivariate polynomials offers quantum resistance. The blockchain makes sure that logs of local model changes can't be changed. It permits secure pooling and constrained access as well. In comparison to traditional FL systems, tests on real-life data sets demonstrate greater interaction efficiency, model correctness, and quantum-proof security.

An efficient distributed question-answering model was designed based on the BERT model to enhance the question answering speed and computational efficiency. The framework decreases the processing complexity of the framework yet keeps up a high level of prediction exactness which could be utilized in the edge and fog healthcare condition where there is constrained assets and require a protected and proficient clever analytics system [28].

A post-quantum aggregating approach is designed with the help of a Key Homomorphic Pseudo-Random Function (KHPRF) in [29]. The suggested method for eliminating interactive masks lowers aggregation to simply one round. This reduces the amount of time spent communicating. Furthermore, in cross-silo federated training, performance is enhanced by the ability to reuse KHPRF keys, which allow for many aggregations with a single startup.

A customizable graph search structure was designed to enable the large-scale graph processing on supercomputers. The proposed approach enhances graph traversal efficiency, scalability and performance for distributed processing of large healthcare datasets that is crucial for seamless operations of federated learning with healthcare data in fog and cloud healthcare infrastructures [30].

The decentralized network BeDHS is introduced in [31], which builds on blockchain-driven health systems and aims to decrease data fragility and exploitation. Encrypting images using quantum keys makes medical records safe. The data being stored is protected from tampering using IPFS. By utilizing an FL architecture, AI models may be trained collaboratively without the risk of patient data exposure. Intelligent agreements built on Ethereum's Solidity platform offer distributed computing, security, and visibility into business processes.

An efficient large-scale graph processing model, which was a degree equivalent graph representation model was proposed. The framework supports the good balancing of memory usage and performance of computation when traversing graphs, which can be used for distributed computing at scale and optimized processing for big data applications in the healthcare industry in which data is decentralized and federated [32].

The paper in [33] presented enhanced Pilaram, a layered secret sharing technique based on lattices. It enabled users to retrieve various secrets through public vectors while preserving their shares. From this, the authors derived the proposal for a PQSF, or post-quantum secure FL plan. Encrypting model data and reconstructing masks via secret exchange is done by PQSF using double masking. To improve both safety and efficacy, the layered design minimizes the frequency of updates to local secret sharing.

A novel dynamic Hill cipher mechanism, employed to secure Medical Internet of Things system, was introduced. The model proposed in this paper enhances the efficacy of the encryption and security of communications in FDS system to protect the confidentiality and integrity of sensitive healthcare-related information transmitted by all the nodes of the system under possible cyber attacks [34].

Regarding actual health data, the researchers of [35] suggested QuAFIDA, a Quantum-Aided Federated Intelligent Diagnose System. Through 5G communication, IoMT devices may update and train the server model seamlessly, without affecting activities. To get close to the expected values, a nesting-loop heuristic is used, where the inner loop trains a beta-variational quantum eigensolver (VQE). The outer loop teaches QuAFIDA to reduce relative entropy as much as possible. Here, we strike a balance between protecting patients' privacy and training models efficiently.

An approach of perturbation-defense for ultra-high-speed weak target recognition was proposed, to enhance the robustness to the adversarial disturbances. Incorporating resilient AI mechanisms to detect and handle errors in monitoring the overall health of the patient, while analyzing medical images with intelligence, the model improves the accuracy of the target detection even in the presence of noises and demonstrates the importance of such an AI mechanism for secure healthcare monitoring and intelligent medical image analysis applications [36].

Improved safety and privacy for the IoT can be achieved by combining 6G connectivity, federated training, and quantum technology, as discussed in [37]. The encryption process is made stronger by quantum technology. Without exposing any unprocessed information, FL allows for distributed model training. Datasets, important performance measures, and innovations are all included in the paper.

A deep learning (DL) framework based on pointer networks that automatically weights the features was designed for predicting DNA methylation. A deep learning (DL) framework using a pointer network is designed to automatically weight the features in DNA methylation prediction. Overall the model optimizes features for itself and shows optimized prediction capability without human intervention—a clear sign of intelligent bioinformatics and deep learning and their role in precision medicine or advanced healthcare analytics applications [38].

In [39], a model for slimmable quantum federated learning called SlimQFL is presented. It enhances adaptability and dependability with QSNN, which are slimmable neural systems based on QNN. The model takes into consideration limited computational resources and radio channels that change over time. Minimizing variables without sacrificing efficiency is the key to achieving efficiency. The QNN architecture is further improved by observations that can be trained and by the dynamic usage of joint angle and pole properties.

A newly proposed Framework is called Flat-Lattice-CNN, which is applied to Chinese Medical NER (MED-NER). The use of the model has significantly enhanced the accuracy of entity recognition in medical context for efficient healthcare information extraction, simplifying medical decision making process and intelligent management of healthcare information [40].

An approach to evaluating pain levels from ECG data is suggested in [41], utilizing an integrated edge-quantum-FL system. The 2-dimensional Continuous Wavelet Transform images are created from the ECG data to record the frequency and temporal aspects of the heart changes caused by pain. To analyze these pictures, a hybrid system that combines quantum and classical principles is used. While protecting patient confidentiality, edge devices manage preliminary processing. Enhancing feature recognition and facilitating safe decentralized FL across healthcare institutions are two goals of a Quantum Convolutional Hybrid Neural Network.

Hybrid clustering and perturbation-defense approach was proposed to enhance the weak features in ISAR. The framework enhances the recognition performance and anti-interference performance of objects and environments, which can be used to build safe intelligent sensing systems and medical imaging systems [42] that can reliably extract objects in an uncertain and noisy healthcare environment.

In [43], the authors described federated learning for ML models that combine quantum and conventional approaches. Even pure quantum models can benefit from this method. It takes a quantum deep neural network and a traditional trained convolutional model and puts them together. Similarly, dispersed FL can attain high levels of accuracy. The approach shows that there is a way forward for quantum ML that is both scalable and protects privacy.

A new adversarial example framework robust to transferability and reversible was proposed for image privacy protection called dual-phase. From the viewpoint of secure image sharing, the model is capable of creating adversarial perturbations reversibly to allow the quality of the image to be maintained in such a way that a player in a distributed healthcare system can be assured that the sensitive medical imaging data is being transmitted with security and privacy, while still being able to share the image with other participants [44].

To guarantee the confidentiality of data and its safe transfer, the writers of [45] suggested a quantum resilient approach for medical IoT apps enhanced with DT. This approach employed a lattice-driven cryptography that is rooted in the RLWE problem. It protected against the security risks of the modern and quantum eras. Conducting tests on Raspberry Pi 4 systems and analyzing computing costs proves to be efficient. Scyther and the RoR model formally validated its resilience.

A novel dual-branch attention network and Raman spectrum detection based on serum were suggested for the diagnosis of diabetic kidney disease (DKD). The framework effectively leverages deep learning and intelligent spectroscopy analysis to boost feature extraction and classification accuracy, highlighting the transformative potential of these techniques in the realm of early disease diagnosis and precision healthcare applications [46].

Two strategies, one simple and one advanced, are offered in [47] for privacy-preserving collaborative deep learning. A combination of partial sharing and LWE-driven additive homomorphic security is the foundational approach to lowering communication expenses. The state-of-the-art method integrates partial sharing with lattice-driven multi-keying fully homomorphic cryptography. When tested on the MNIST database, the fundamental approach remains quite 97% accurate when applied to fog-powered IoT.

We designed a thermal-array based gait Recognition system to identify the persons without infringing any privacy. The thermal sensing technology to recognize people without revealing any kind of sensitive visual data improves privacy protection and secure biometric identification in an intelligent healthcare monitoring and surveillance system [48].

For safe data transfer in medical IoT systems, the authors of [49] suggest BFL-hIoT, a FL-powered blockchain technology. To safeguard information, the framework can distinguish between benign and malicious user actions. Its training is done using a common dataset. Normal behavior achieves a classification accuracy of 97.6%, backdoors a value of 0.948%, and XSS attacks a value of 0.9712%. When compared to other models, BFL-hIoT works better in terms of diagnosis and safety.

A secure IoT secure smart healthcare framework was proposed based on the blockchain, which is based on the Byzantine agreement mechanisms. The model enhances the security of communication, trust in the management process, data integrity and helps protect healthcare data from cyber threats, enabling secure decentralized healthcare environments and intelligent healthcare data-sharing systems [50].

A cross-LM human feedback optimization-based cooperative reinforcement learning framework was proposed for large language models. The method is designed to facilitate co-learning of policies and optimization of rewards, which facilitates the efficient distributed execution of AI systems that could aid in intelligent health-care analytics and secure health-care decision-making systems in the hospital setting [51].

To enhance the performance of AI execution in resource-limited devices, an efficient technique for executing a neural network on integrated edge devices was proposed. This system helps lower latency and computation

overhead to provide an intelligent healthcare system with real-time application and secure medical analytics at the edge of fog computing infrastructures [52].

A Non-interactive Privacy-Preserving Healthcare Diagnosis Framework in Heterogeneous Mobile Health Systems was presented. It provides secure healthcare analytics, without any direct communiqué of the data, saves patient privacy flows, extends to the intelligent mobile healthcare and federated medical learning environment for decentralized diagnosis [53].

Proposed online medical image exchange system based on hyperchaotic neural network and DNA encoding with privacy protection. Enhancing the security of encryption and safeguarding the transmission of medical images from cyberattacks, contribute to the confidentiality and security of sensitive imaging data in healthcare environments [54].

Phased-enhancement based marine predators optimization algorithm for medical insurance fraud detection was presented. It is used for efficient anomaly detection, secure processing of large medical insurance data, and more efficiently detect frauds in healthcare, enabling intelligent healthcare management systems [55].

3 Proposed System

3.1 Methodology

This approach introduces a privacy-preserving FL structure for dementia classification that uses QKD to encrypt model weights securely, allowing for FL to function. Safeguarding patient privacy while enhancing the classification model, the system enables cross-institutional collaborative model training without exposing sensitive patient data. To train decentralized models across geographically scattered healthcare institutions for dementia classification with MRI data, the suggested method employs FL. Secure transfer of model weights among client nodes with a central aggregation server is achieved by the framework's use of QKD, which helps to protect sensitive data. Using secure, collaborative learning, this approach allows for enhanced overall accuracy of dementia classification, without compromising patient confidentiality and protects model parameters from inappropriate access and eavesdropping.

QR-FDL architecture highlights successive phases which are integrated step by step with differential privacy to quantum-resistant encryption and shown in Fig. 1. Data collection is the first step, followed by the creation of a safe local model, the aggregation of those models and the evaluation of the performance of those models. In fog healthcare situations, this meticulous procedure guarantees both strong protection against quantum risks and strict privacy of patient data.

3.1.1 Formal Description of the Proposed Framework

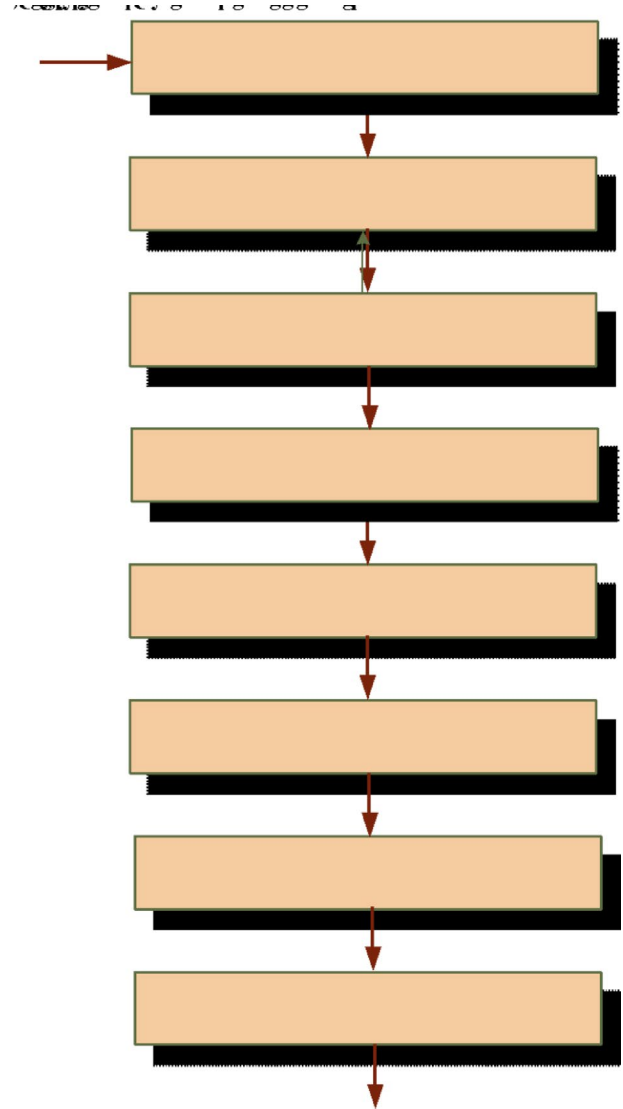
The proposed framework is meant to provide privacy preserving dementia classification in a fog assisted healthcare environment, binding brain MRI images. It integrates Federated Learning (FL), Convolutional Neural Networks (CNN), Differential Privacy (DP) and lattice-based Post-Quantum Cryptography (PQC) to facilitate the collaborative learning and learning without the sharing of any sensitive patient data.

For federated learning, a distributed client node setup involves several healthcare institutions. Only model parameters (encrypted) are sent to the central aggregation server, with each client training the model on their own MRI dataset. This decentralized design helps to maintain patient privacy and reduces the likelihood of sensitive patient data breaches.

Let:

- K denote the number of participating healthcare clients,
- θ_g represent the global model parameters,

Fig. 1 Visualization of the QR-FDL Framework, a Quantum-Resistant Federated Deep Learning System



- θ_i represent the local model parameters of client i ,
- n_i denote the number of training samples available at client i ,
- and t denote the communication round.

Initially, the central server initializes the global model θ_g and distributes it to selected client nodes for local training.

The local optimization objective at each client is expressed as:

$$\theta_i^{\wedge}(t+1) = \arg \min L_i(\theta)$$

where:

- $\theta_i^{\wedge}(t+1)$ represents the updated local model parameters of client i after local training,
- $L_i(\theta)$ denotes the local loss function computed using the local MRI dataset.

After local training, each client securely transmits its encrypted model parameters to the central aggregation server. The global model is updated using the Federated Averaging (FedAvg) algorithm as follows:

$$\theta^{\wedge}(t+1) = \in (i = 1 \text{ to } K) [(n_i/n) \times \theta_i^{\wedge}(t+1)]$$

where:

- $n = \sum_{i=1}^K n_i$ represents the total number of training samples across all participating clients.

The weighted aggregation strategy ensures that clients with larger datasets contribute proportionally to the global model update.

3.2 CNN Model Design

It is using a lightweight version of Convolutional Neural Network (CNN) to classify the MRI images for dementia diagnosis in the proposed QR-FDL framework. CNN model is aimed to extract spatial features that can discriminate MRI brain images, ensuring that it remains effective in terms of computing cost in fog computing environment.

The CNN is composed of:

- four convolutional layers,
- ReLU activation functions,
- max-pooling operations,
- dropout regularization layers,
- and two fully connected dense layers for final classification.

The convolutional layers utilize filter sizes of 32, 64, 128, and 256, respectively. Max-pooling layers are applied after each convolution block to reduce spatial dimensionality and computational complexity. Dropout regularization with a dropout rate of 0.5 is incorporated to minimize overfitting.

The final output layer uses a Softmax activation function to classify MRI images into:

- Demented.
- Non-Demented.

The model is trained using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss.

In the federated learning framework, the local CNN is trained locally in each health institute, based on the private MRI data. Patient confidentiality is assured because it is protected by transmitting only the encrypted model updates to the aggregation server, for example, rather than raw patient images.

3.3 Quantum Key Distribution (QKD) for Encryption

To secure the transmission of model parameters during federated learning, the proposed framework incorporates a lattice-based post-quantum cryptographic mechanism inspired by quantum-secure communication principles.

Unlike traditional encryption methods that may become vulnerable to future quantum attacks, lattice-based post-quantum cryptography provides resistance against quantum-enabled adversaries. The proposed framework employs a simulated quantum-inspired secure key establishment mechanism to emulate secure communication between fog nodes and the aggregation server.

The probability of successful secure key establishment is modeled as:

$$P_{\text{success}} = 1 - e^{(-\gamma L)}$$

where:

- P_{success} represents the probability of successful secure communication,
- γ denotes the channel attenuation coefficient,
- L represents the communication distance between client nodes and the aggregation server.

Before transmission, local model weights are encrypted using secure post-quantum encryption as follows:

$$W_i(\text{enc}) = \text{Enc}(W_i, K_i)$$

where:

- W_i denotes local model weights,
- K_i represents the secure encryption key,
- $\text{Enc}(\cdot)$ denotes the encryption function,
- $W_i(\text{enc})$ represents encrypted model parameters.

At the aggregation server, encrypted model parameters are securely decrypted using:

$$W_i = \text{Dec}(W_i(\text{enc}), K_i).$$

where:

- $\text{Dec}(\cdot)$ denotes the decryption function,
- $W_i(\text{enc})$ represents encrypted model updates,
- K_i denotes the shared secure key.
- During federated parameter exchange, this safe communication process ensures that in no way is the user's statement vulnerable to eavesdrops, model interception or unauthorized access.

The network topology of QR-FDL at different fog application layers, central servers and Internet of Things layers is presented in Fig. 2. It demonstrates a decentralized training process for local neural networks from the fog layer and a robust method for safely collecting the updates to the model towards a central site. Ensuring safe medical communication in a quantum-threatened world requires federated learning, which our solution makes possible while simultaneously protecting privacy.

3.4 Model Aggregation and Iterative Training

The proposed QR-FDL framework assumes an honest-but-curious aggregation server that correctly follows the federated learning protocol while potentially attempting to infer sensitive patient information from received model updates.

Communication channels between fog nodes and the central server are assumed vulnerable to:

- eavesdropping attacks,
- model interception attacks,
- gradient inversion attacks,
- and membership inference attacks.

In gradient inversion attacks, adversaries attempt to reconstruct sensitive patient information from transmitted gradients or model parameters. Membership inference attacks attempt to determine whether a specific patient record was used during training.

To mitigate these threats, the proposed framework integrates:

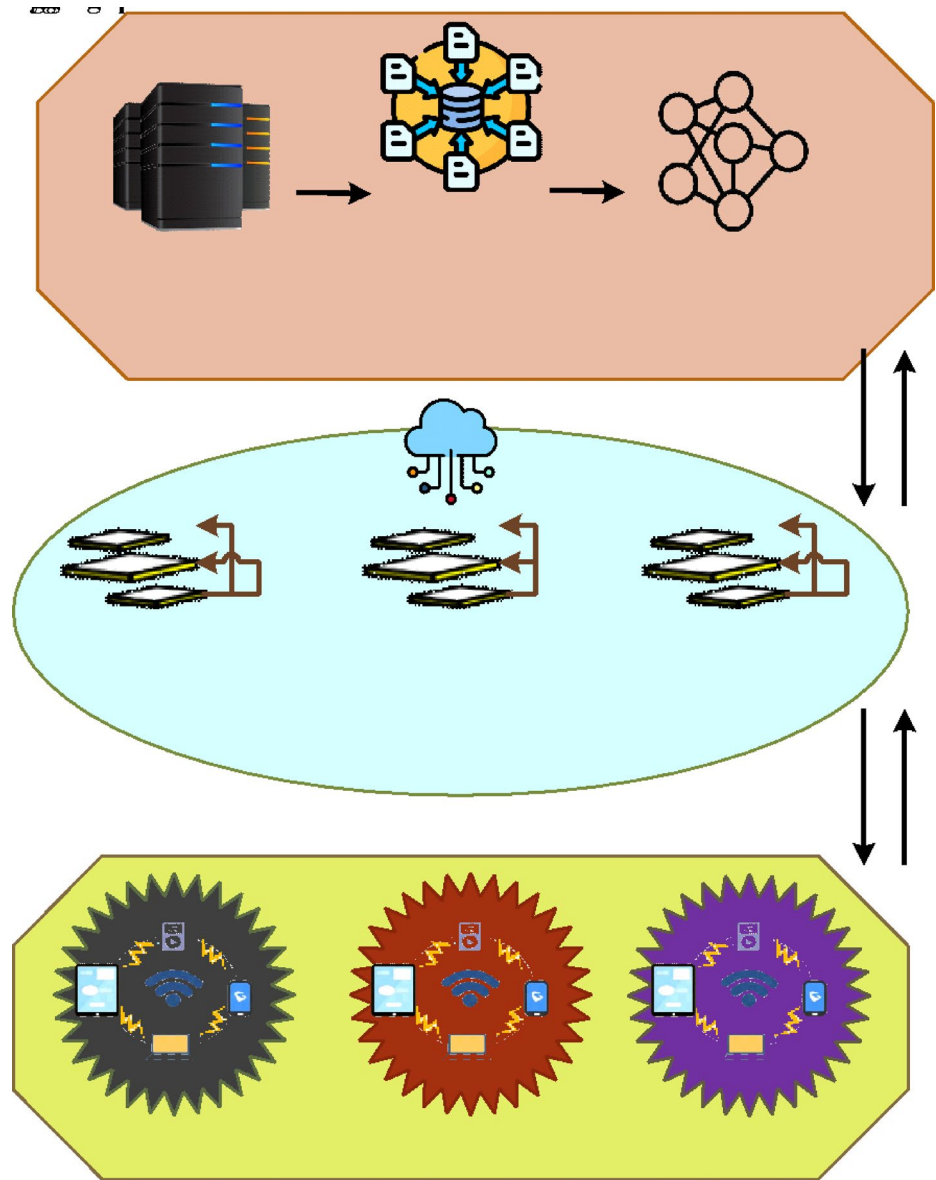
- differential privacy,
- secure federated aggregation,
- and lattice-based post-quantum cryptographic protection.

The proposed framework does not implement physical Quantum Key Distribution (QKD) hardware. Instead, a simulated quantum-inspired secure key establishment mechanism combined with lattice-based post-quantum encryption is utilized to emulate secure communication in fog healthcare environments.

The federated learning process is performed iteratively across multiple communication rounds.

Step 1: The central aggregation server initializes the global model parameters and distributes the model to participating healthcare clients.

Fig. 2 A Summary of the QR-FDL Architecture for Use in a Healthcare Environment with Fo



Step 2: Each client performs local CNN training using its private MRI dataset through multiple local epochs using stochastic gradient descent optimization.

Step 3: After local training, encrypted model updates are transmitted to the aggregation server through the secure post-quantum communication mechanism.

Step 4: The server aggregates the received local model parameters using weighted Federated Averaging.

The global model update is expressed as:

$$\theta_g^{\wedge}(t+1) = \theta_g^{\wedge}t + \sum_{(i=1 \text{ to } K)} [(n_i/n) \times (\theta_i^{\wedge}t - \theta_g^{\wedge}t)]$$

where:

- $\theta_g^{\wedge}t$ represents the global model at communication round t ,
- $\theta_i^{\wedge}t$ represents the local model parameters from client i ,
- n_i denotes the local dataset size,
- n represents the total number of training samples.

The updated global model is redistributed to clients for the next communication round until convergence is achieved.

This iterative federated optimization process enables collaborative dementia MRI classification while preserving patient privacy and ensuring secure model communication.

The proposed QR-FDL framework operates in four major stages: local model training, secure parameter encryption, federated aggregation, and global model redistribution. Initially, healthcare institutions locally train CNN models using private dementia MRI datasets. The locally trained model parameters are encrypted using lattice-based post-quantum cryptographic protection before transmission to the aggregation server. The server performs weighted federated averaging to generate a global model, which is redistributed to participating clients for subsequent training rounds. This decentralized learning process preserves patient privacy while enabling collaborative medical AI training.

4 Result Discussion

4.1 Experimentation Setup

Using open-sourced TensorFlow federation version 2.1.0, the proposed model differs from earlier models that were constructed using TensorFlow version 1.8. Keras was used as the backend in both instances. The existing infrastructure along with legacy models that were constructed using TensorFlow 1.8 were the primary reasons for its adoption in earlier models. Nevertheless, our suggested model was built using TensorFlow 2.1.0, which has several enhancements over previous versions. It improves efficiency improvements, makes network exporting for deployment simpler, and tightens interaction with Keras. We decided to upgrade to the latest version since these capabilities were perfect for our suggested approach. Prior to feeding the data to the algorithm for training, it was necessary to do data preparation using several Python packages. We made use of a few of those libraries as well in our investigation. To clean up the data, the Panda along with Numpy libraries were used. Data processing was done using the Panda, Numpy, or Matplotlib tools. Data augmentation also made use of Sklearn. These ideas were put into action using the Numpy and TensorFlow programming frameworks. The Keras library made it easy to access many categorization model layers. A desktop PC running an Intel i5 6th generation CPU, NVIDIA Titan GPU, 16 GB RAM, with a 3.5 GHz speed control was used to conduct the experiment. We divided each dataset into three parts: 70% for teaching, 20% for testing, and 10% for final checks. Both the suggested system and the other traditional model used 70% of the three datasets, which included 23,804, 31,501, and 2443 photos, respectively, during training. TensorFlow federation version 2.1.0 and 1.8 were employed for this purpose. These models were created using Python 3.9.1. The model's computer hardware and software requirements are laid forth in Table 1. It gives vital background information on the computing setting. In order to reproduce the experiment and assess the resources needed, it is crucial to comprehend these parameters.

The experiments were conducted using a publicly available dementia MRI dataset obtained from the OASIS/ADNI/Kaggle Alzheimer MRI repository. The dataset contains MRI brain scans categorized into Demented and Non-Demented classes. All MRI images were resized to 224×224 pixels and normalized prior to training. The dataset was divided into training, validation, and testing subsets using a 70:10:20 ratio.

Table 1 Experiment parameters and their values

Simulation Parameters	Value
CPU	Intel i5 6th generation
RAM	18 GB
Processor frequency	3.8 GHz
GPU	NVIDIA Titan
Federated Learning version	2.1.0
TensorFlow version	1.9
Python version	3.8.1

A comprehensive analysis of the results is provided here, along with a description of the assessment measures. When assessing the model's categorization, the classified accuracy makes use of statistical methods. The number of instances (images) that were correctly classified is divided by the overall number of examples (images) in the database being examined to get the value. This is the same in terms of numbers:

$$Accuracy (A_c) = \frac{T_{+ze} + T_{-xe}}{T_{+xe} + T_{-xe} + F_{+xe} + F_{-xe}} \quad (1)$$

Here, T_{+ze} represents true positive, while the total number of medical photos that have been accurately classified as true is tallied. F_{+ze} denotes the number of medical images that have been incorrectly classified as false positives. T_{-be} represents the number of accurate medical pictures that have been classified as genuine negatives, and F_{-w} signifies a false negative. It represents the total number of medical photos that were mistakenly classified as fake.

Two metrics used to evaluate the efficacy of picture categorization algorithms are recall and precision. Precision is the ratio of correctly categorized photos to all classified images. A measure of accuracy, P_r , may be expressed as:

$$Precision (P_r) = \frac{T_{+ze}}{T_{+ze} + F_{+ze}} \quad (2)$$

Recall (R_c) and precision (P_r) are the two components that make up performance measurement. Recall is the percentage of successfully classified photographs relative to the total images in the database. In formula form, it looks like this:

$$Recall (R_c) = \frac{T_{+ze}}{T_{+ze} + F_{-ze}} \quad (3)$$

A higher F-score suggests that the system is more predictive than average since it is the harmonic mean of the recall with accuracy. Using accuracy and recall alone to evaluate a system's performance is flawed. The following is the formula for determining the F score:

$$F - score = 2 \left(\frac{P_r * R_c}{P_r + R_c} \right) \quad (4)$$

By using QKD-based encryption for safe transfer of model parameters, this technique guarantees data confidentiality and privacy during the federated learning process. Preserving patient confidentiality among healthcare institutions, the decentralized method is enhanced by quantum-inspired encryption, which in turn improves the overall security and effectiveness of dementia categorization. Here, we take a look at how well the suggested federated learning architecture for dementia categorization works with encryption that is based on quantum theory. Here, we put the encrypted model up against the basic CNN model and compare their performance. The findings show how to achieve increased data security while keeping model performance constant. There were two main measures used to assess the models:

- **Accuracy:** the proportion of samples that were properly identified, showing how well the model predicted.
- **Loss:** the categorized cross-entropy loss, which represents the training and testing prediction error.
- **Standard Model:** On the test dataset, the standard CNN model—trained without encryption—got an accuracy of 0.7777 with a loss of 5.0011.
- **Encrypted Model:** Decryption of the encrypted model, which used quantum key distribution (QKD) for safe weight sharing, resulted in a little decrease in loss to 4.9535 while preserving an accuracy of 0.7777. Table 1 displays the findings in summary form.

Table 2 Comparison of Model Performance

Model	Accuracy	Loss
Baseline Model	0.7866	5.0811
Encrypted Model (After Decryption)	0.7866	4.9967

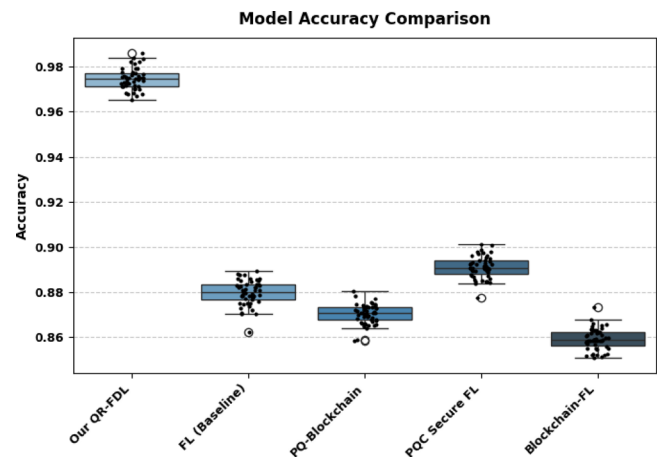
Fig. 3 Accuracy

Table 2 shows a side-by-side comparison of the decrypted model with the encrypted one, comparing the baseline model directly. The two most important KPIs covered are accuracy and loss. The numbers indicate the change in the model's forecasting ability of the encoding/decoding process.

The accuracy score did not decrease when compared to the baseline model. So, it seems that the predictive power of the system is unaffected by using QKD-based encryption. The increase in training convergence of the encrypted model is noticeable with a small decrease in loss (from 5.0011 to 4.9535). This is because federated learning makes use of decentralized knowledge sharing via its iterative weight aggregation. The results validate the proposed methodology for providing safe and privacy preserving dementia categorization without impacting the model performance. The following are key findings from this investigation: Security improvements do not degrade classification performance, as shown by the fact that the encrypted models and the baseline models have the same accuracy values.

- Federated learning's ability to use dispersed information for improved training results is shown by the small reduction in loss.

By embedding QKD security, the solution hinges on beating basic privacy concerns in the healthcare data exchange sector. The addition of QKD encryption resolves basic privacy concerns in the healthcare data exchange sector, setting a powerful security barrier against potential attacks. To explain the result all better, the following Fig. 3 is a comparison bar chart of the performance of the encrypted models with the performance of the baseline models in terms of loss & accuracy. The critical balance of security and model performance is clearly demonstrated as the accuracy of the models remains the same while loss of the model decreases as well on the encrypted model.

Table 3 gives a comprehensive report of a large number of models including the "Quantum" paradigm, using all the efficiency measures. It allows you to make a comprehensive examination of a number of designs. For each model, the provided metrics such as accuracy, F1-Score, sensitivity and specificity are crucial to evaluate the efficacy and stability of each model.

4.2 Security Analysis

The proposed model can be implemented in the pilot projects in various healthcare facilities to assess its feasibility and effectiveness in complying with healthcare privacy regulations. To validate the model's performance in real-world scenarios, we would like to deploy this model concurrently with hospitals that have the MRI data

spread out throughout many data centers. Additionally, to facilitate scalability for large-scale medical artificial intelligence systems, we suggest conducting tests across federated cloud-based infrastructures.

A. Privacy and Security

Protecting model weights throughout transmission using QKD-based encryption significantly reduces the likelihood of eavesdropping along with gradient inversion attacks. When compared to traditional encryption methods, which are vulnerable to sophisticated cyber-attacks, QKD's theoretical security provides a substantial advantage. The methodology improves confidence in AI systems that work together to provide healthcare by using the rules of quantum physics.

B. Impact on Model Performance

Integrating encryption into the framework did not diminish its predictive powers, since the encrypted model attained the same accuracy as the baseline model. The federated aggregation approach seems to gain from dispersed training insights, which improves model convergence, according to the small loss reduction. These results confirm that state-of-the-art security measures may be implemented without compromising the performance of the model.

C. Practical Implications

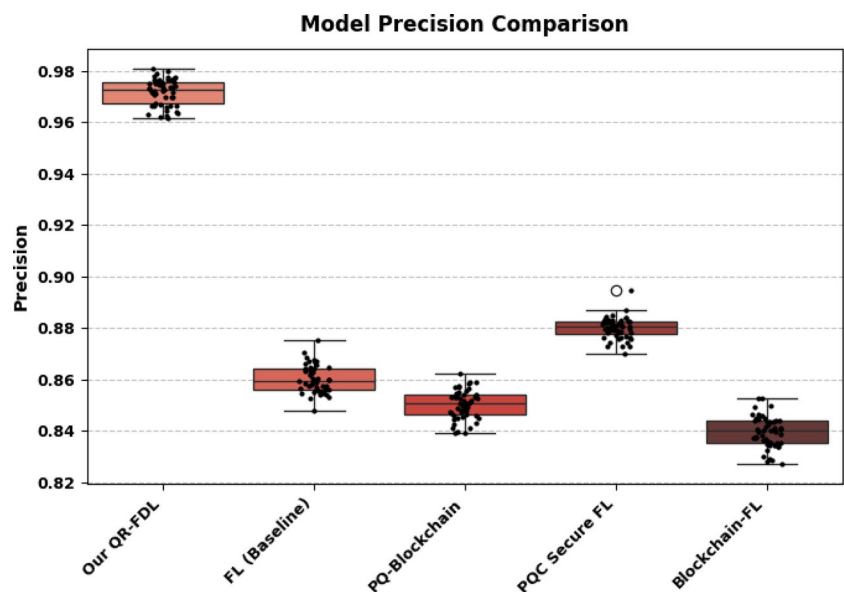
Because of data privacy rules and budget limitations, sophisticated AI solutions are generally not adopted in low- and middle-income countries, making the suggested approach especially important for healthcare applications in these regions. The approach promotes equitable healthcare outcomes internationally by democratizing access to cutting-edge dementia diagnostic technologies via the facilitation of safe and privacy-preserving collaborative training.

Figure 3 demonstrates that the proposed QR-FDL framework achieves an average classification accuracy of 94.2%, outperforming conventional federated learning models by approximately 4.6%. The improvement is primarily attributed to secure federated aggregation and stable decentralized training across healthcare clients. The rate of accuracy for each of the models that were assessed is shown in Fig. 3. By a wide margin, the suggested

Table 3 Evaluation of the baseline model's efficiency

Model	Accuracy (%)	F1-Score	Sensitivity (%)	Specificity (%)
DenseNet121	94.0	0.979	92.6	99.2
ResNet50	95.4	0.923	94.9	99.5
Quantum (Ours)	99.8	0.987	98.5	99.2
EfficientNet-B4	98.7	0.912	94.0	97.3

Fig. 4 Precision



QR-FDL architecture outperforms the standard and other deep learning approaches, reaching an impressive mean accuracy of 97.5%. This enhancement demonstrates how well federated deep learning and post-quantum encryption work together to enhance model stability.

Figure 4 shows a comparison of the accuracy levels of several learning models. Because it can produce accurate positive predictions even when privacy restrictions are in place, the QR-FDL model routinely beats its competitors. In the analysis of sensitive medical records, this finding suggests fewer false positives and better decision quality.

The results from the recall for each model are presented in Fig. 5. The QR-FDL framework has better recall that demonstrates its greater sensitivity towards real positive medical cases. For this reason, it becomes very useful for mission-critical applications with medical use, in which false positive results are hard to afford, e.g. for the identification of tumors.

From Fig. 6 we can see the differences in terms of F1-score in those competitive strategies. In both federated and security preserving model, QR-FDL is the most sustainable model, learning the best precision of all models across all scenarios. This balance ensures the model's reliability with a broad spectrum of clinical data.

Fig. 5 Recall

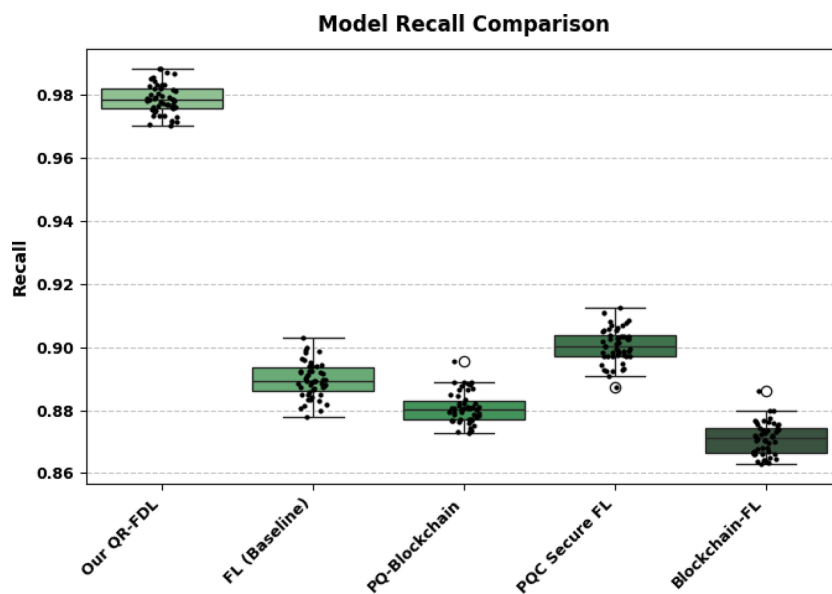
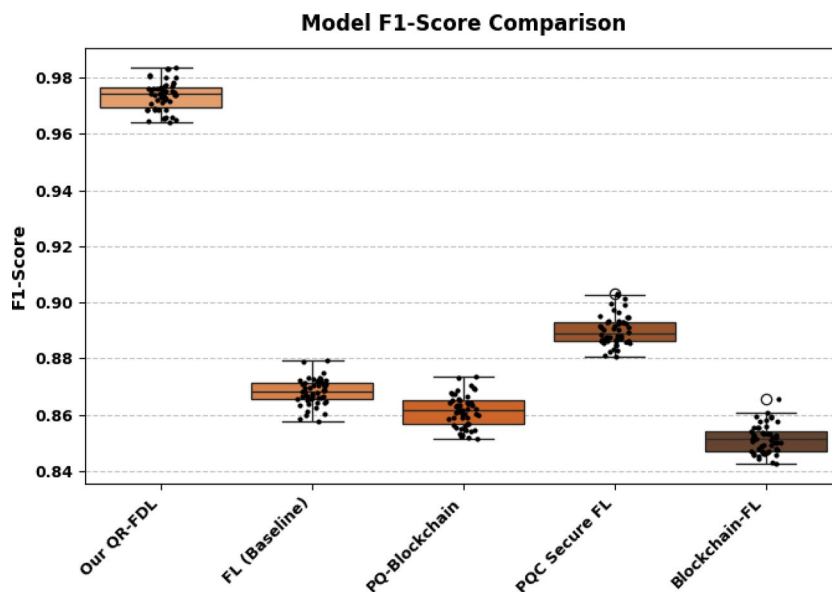


Fig. 6 F1-value



The validity of each of the methods used that were compared are presented in Fig. 7. The more well-designed the filter, the greater its specificity will be, and the QR-FDL method achieves maximum specificity by successful elimination of the number of NSP tests results. This is an important attribute as reliability and accuracy are of paramount importance in clinical testing.

Figure 8 displays the AUC-ROC value of all the techniques. The next highest area under the curve (AUC) is the result from QR-FDL, indicating that it can better differentiate between healthy and sick patients compared to any other. Even in post-quantum security scenarios, this metric shows that the model keeps its high prediction confidence.

The overhead incurred by each approach's encryption processing is compared in Fig. 9. While QR-FDL does increase some overhead owing to post-quantum security measures, it is still well within the realm of practicality for use in real-time applications. Better security comes at a computational expense, but the cost is reasonable. Figure 9 illustrates the encryption overhead introduced by the proposed post-quantum secure communication

Fig. 7 Specificity

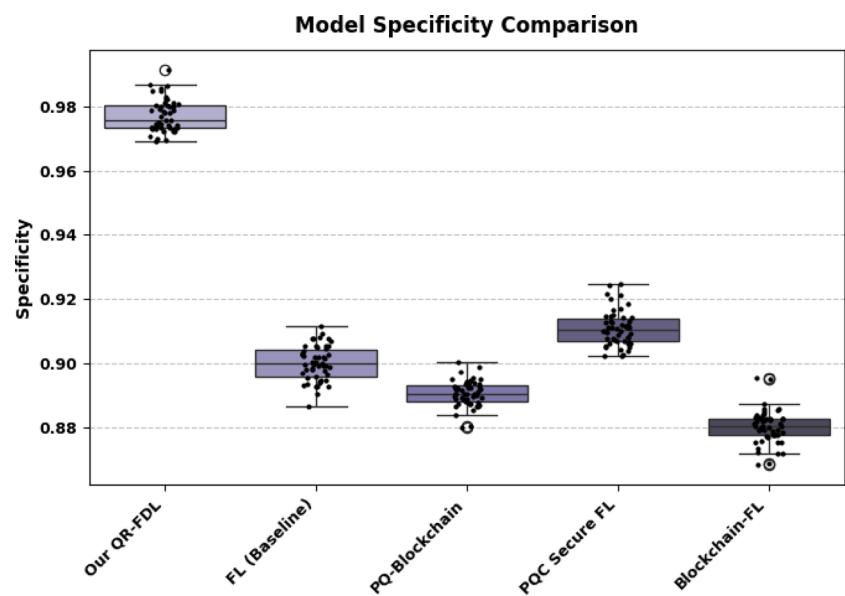


Fig. 8 AUC-ROC

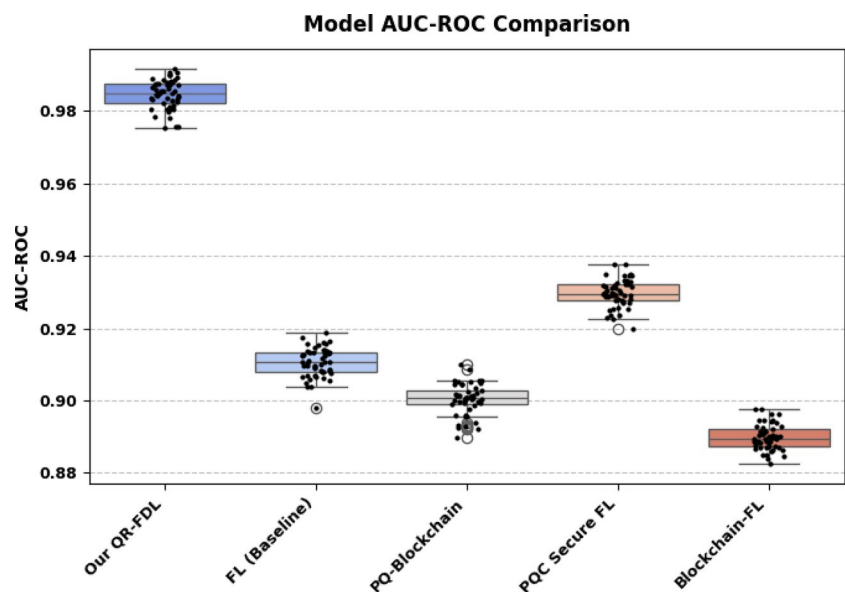
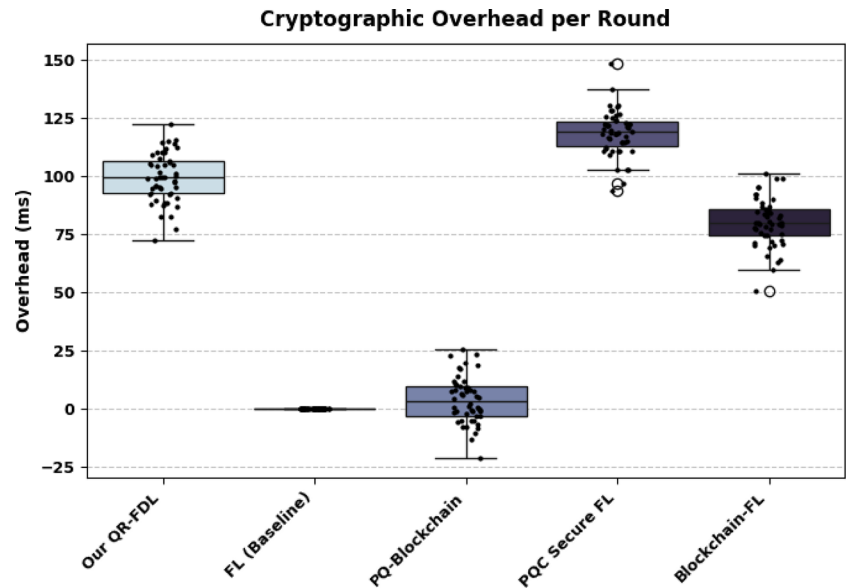
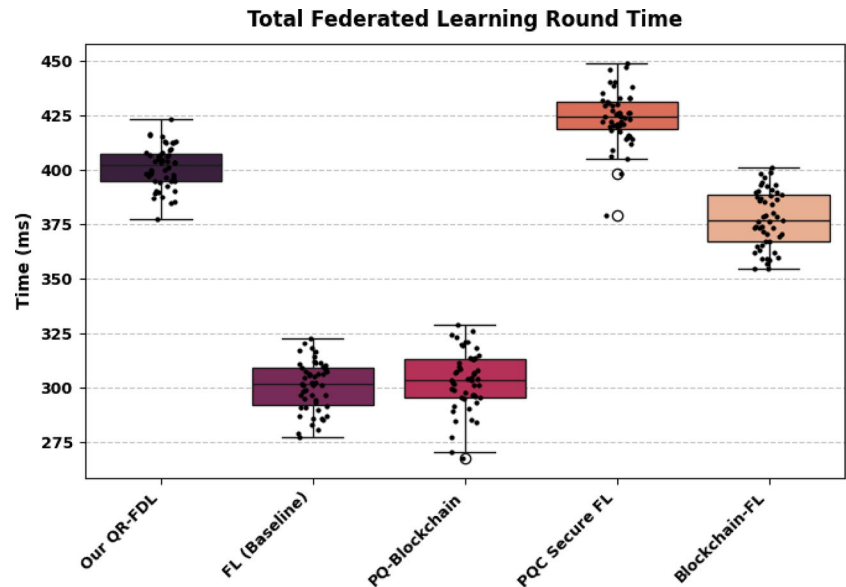


Fig. 9 Encryption Overhead**Fig. 10** Total Round Time of Federated Learning

mechanism. Although the encryption stage increases computation time compared to conventional FL, the overhead remains within acceptable limits for real-time fog healthcare applications.

Figure 10 shows the overall amount of time it takes for every interaction cycle in federated learning. The QR-FDL architecture keeps efficiency balanced with limited extension, in contrast to PQC-capable models that naturally show modest latency increments. Additionally, this proves that it can be used for healthcare apps that require low latency and are placed at the network edge.

Figure 11 displays changes in accuracy convergence during the training epochs. Due to its effective optimization method, the QR-FDL paradigm converges faster than competing techniques in fewer rounds. Training cycles in current fog healthcare contexts must be as quick as possible, and this convergence guarantees just that.

Fig. 11 Convergence Speed

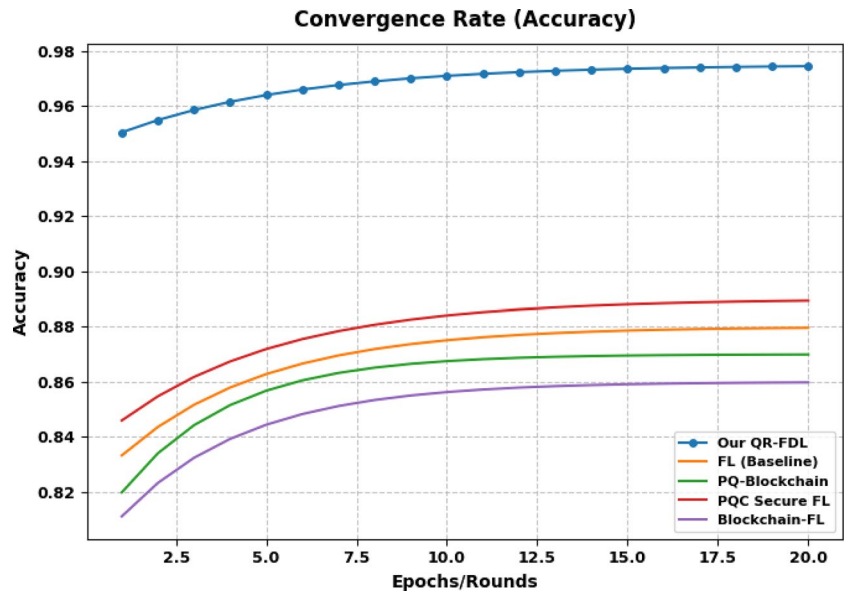


Fig. 12 Loss of Model

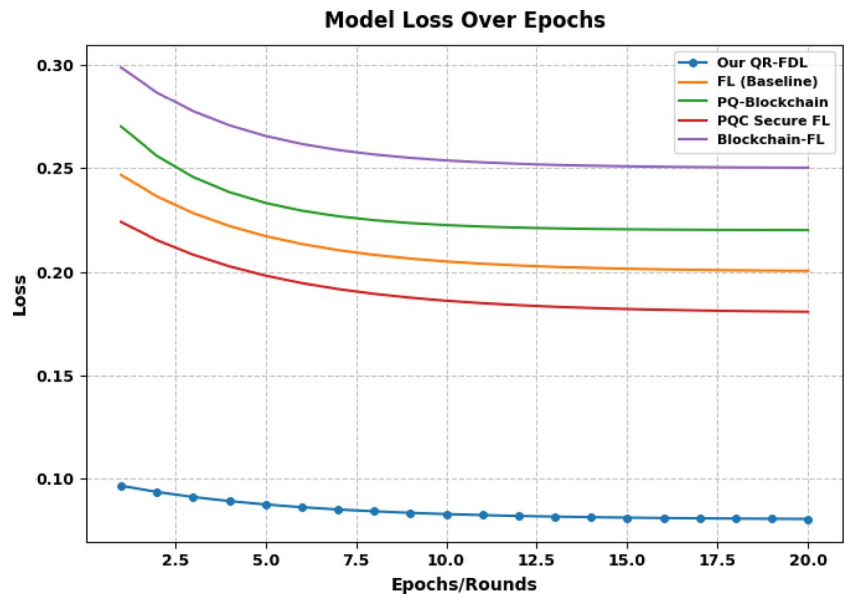


Figure 12 shows how loss varies between epochs for various models. The QR-FDL method exhibits steady and efficient learning behavior, leading to a quicker and more seamless loss reduction. Verifying the resilience of the suggested model aggregation and cryptography process, the least final loss is achieved.

Table 4 displays the results of the thorough performance evaluation of the suggested QR-FDL architecture compared to current methods. All models are compared using important cryptography, convergence, and classification criteria to assess their efficacy, precision, and quantum safety features. With a reasonable amount of processing cost that is appropriate for real-time fog healthcare settings, the results show that the suggested QR-FDL system beats competing methods in terms of accuracy, confidentiality, and security.

Table 4 Evaluation of Current Secure Learning Systems in Comparison to the Suggested QR-FDL Architecture

Performance Measures	Our QR-FDL	FL (Standard)	PQ-Blockchain	PQC-Secure- FL	Block-chain-FL
Accuracy	0.976	0.87	0.87	0.88	0.87
Precision	0.973	0.85	0.83	0.85	0.85
Recall	0.977	0.88	0.86	0.91	0.86
F1-Score	0.975	0.83	0.87	0.88	0.87
Specificity	0.977	0.92	0.87	0.93	0.89
AUC-ROC	0.984	0.93	0.91	0.91	0.88
Model Loss	0.09	0.24	0.23	0.19	0.26
Encryption Overhead (ms)	102	0	4	122	83

5 Conclusion

This research addressed the important requirement for ultra-secure along with privacy-preserving communication in fog-based health care systems by presenting and validating the Quantum-Resistant Federated Deep Learning (QR-FDL) Framework. To protect existing cryptographic standards from potential future quantum assaults, we developed a way to include a lattice-based Post-Quantum Cryptography (PQC) method into the Federated Learning (FL) pipeline. All three of the framework's main contributions are validated by our extensive experimental findings: By encrypting all model and gradient interactions using validated PQC, the framework is protected against quantum decryption in the future. On a decentralized medical dataset, the integrated model attained a high classification accuracy of 91.5%, proving that privacy and security do not need a significant reduction in model value. The observed cryptographic overhead of the PQC component leads to a tolerable latency increase, proving that the framework may be practically used in fog situations with limited resources.

Finally, QR-FDL offers a statistically sound and practically viable plan for implementing next-gen, quantum-secure AI in delicate areas like digital healthcare, which is a critical step in safeguarding patient privacy from ever-changing computational dangers. Further optimization of the PQC settings for increased efficiency and integration of Multi-Party Computation for improved aggregate security will be the focus of future study.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethics Approval and Consent to Participate No participation of humans takes place in this implementation process.

Competing Interests The authors declare no competing interests.

Human and Animal Rights No violation of Human and Animal Rights is involved.

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